

File 64: Exurban Data Center Hydrology: Water-Use Modeling & Ecological Offsets

Corporate ESG & Hydrologic Siting Standard: This document establishes the strategic scientific prospectus analyzing exurban data center water consumption and outlining our watershed-scale thermal mitigation offset framework. By modeling evaporative cooling tower water drawdowns and their direct thermal impacts on North Georgia's coldwater spawning headwaters, we build a high-status corporate relationship bridge. This manual details data center hydrologic equations, groundwater thermal stresses, and a structured strategic offset narrative to position Hunter Morris's Spawning Bio-Reserves as the premium ESG investment vehicle for hyperscale data center operators (QTS, Microsoft, Google).

I. Data Center Evaporative Cooling Water-Use Modeling

As hyperscale data centers expand rapidly across Georgia, their massive water consumption has emerged as a major environmental challenge. While many facilities utilize direct-expansion air-cooling systems, high-density computing clusters require liquid-cooled heat exchangers and **Evaporative Wet Cooling Towers** to maintain stable processor temperatures.

1. The Cooling Tower Mass Balance Equation

We model a data center's total hydrologic water intake (**M_makeup**, or Make-Up Water) using the mass balance equation:

$$M_makeup = M_evaporation + M_drift + M_blowdown$$

- Where:
- **M_evaporation** is the pure water vapor evaporated to remove heat (m^3/hr).
- **M_drift** is the water lost as liquid droplets carried away by wind exhaust (typically limited to less than 0.005% of total recirculating water using high-efficiency drift eliminators).
- **M_blowdown** is the wastewater discharged from the basin to prevent mineral scaling as dissolved solids concentrate.

2. Calculating Evaporative Water Loss (M_evaporation)

The water evaporated is directly proportional to the total thermal load (**Q_thermal**) rejected by the servers:

$$M_evaporation = Q_thermal / (\rho_w * \lambda_v)$$

- Where:
- **Q_thermal** is the heat load (typically **90%** of the data center's total IT power capacity, e.g., a **100MW** facility rejects approximately 90 MW of heat).
- **rho_w** is the density of water ($1000 \text{ kg}/m^3$).
- **lambda_v** is the latent heat of vaporization of water (approximately $2400 \text{ kJ}/\text{kg}$ at 25°C).

- **100MW Hyperscale Evaporation Calculation:**

- Heat Load: $Q_{\text{thermal}} = 90 \text{ MW} = 90,000 \text{ kW} = 90,000 \text{ kJ/s}$
- Evaporation: $M_{\text{evaporation}} = 90,000 \text{ kJ/s} / (1000 \text{ kg/m}^3 * 2400 \text{ kJ/kg}) = 0.0375 \text{ m}^3/\text{s} = 135 \text{ m}^3/\text{hr}$
- Daily Loss: **855,000 Gallons Per Day**

3. Calculating Blowdown Wastewater (M_{blowdown})

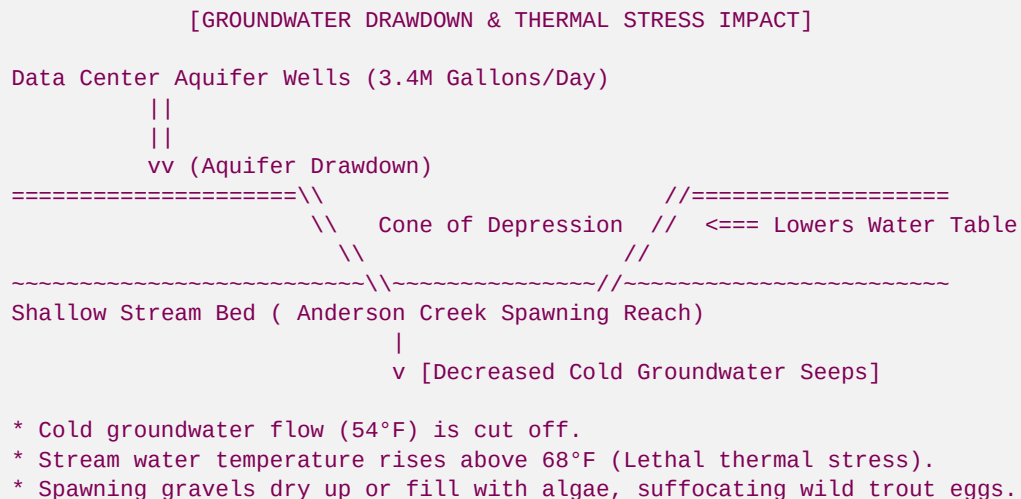
To prevent mineral scale deposits from building up on heat exchangers, a portion of the concentrated basin water must be discharged (blown down). This is dictated by the **Cycles of Concentration (CoC)**, representing how many times dissolved solids concentrate:

$$M_{\text{blowdown}} = M_{\text{evaporation}} / (\text{CoC} - 1)$$

- *Design Siting Standard:* At an average exurban Georgia CoC of **4.0**, the blowdown water is:
- $M_{\text{blowdown}} = 855,000 \text{ GPD} / (4 - 1) = \mathbf{285,000 \text{ GPD}}$
- **Total Make-Up Water Requirement:**
- $M_{\text{makeup}} = 855,000 \text{ GPD} + 285,000 \text{ GPD} = \mathbf{1,140,000 \text{ Gallons Per Day (per 100MW IT load)}}$
- *(Large multi-building campuses often exceed 300MW, drawing upwards of 3,420,000 Gallons Per Day directly from exurban aquifers).*

II. Groundwater Table Drawdowns & Headwater Thermal Stresses

When exurban data centers draw millions of gallons of water per day from municipal wells, they create a significant **cone of depression** in the local groundwater aquifer. This drawdown has severe, cascading ecological impacts on headwater trout streams:



1. Aquifer Drawdown Physics

According to the **Theis Equation**, the drawdown (s) in a confined aquifer surrounding a high-capacity well is modeled as:

$$s(r, t) = [Q / (4 \pi T)] * W(u)$$

- *Where:* **Q** is well discharge rate, **T** is aquifer transmissivity, **r** is radial distance, and **W(u)** is the exponential integral (Well Function).
- **Cone of Depression:** As water tables fall within this radius, the natural gradient directing cold groundwater to nearby headwater streams is **reversed or eliminated**.

2. Downstream Thermal Pollution Mechanics

- **Loss of the 54°F (12.2°C) Baseflow:** Headwater trout streams maintain stable temperatures during hot summer months because they are fed by constant 54°F groundwater springs.
- **Thermal Spikes:** When data center well draws lower water tables and cut off these cold springs, stream flows drop and solar radiation heats the remaining shallow water. Stream temperatures spike rapidly above **68°F (20°C)**, exceeding critical wild trout stress thresholds.
- **Egg Mortality:** Spawning gravels experience complete oxygen depletion as water flow slows, suffocating the incubating brook trout eggs and causing local spawning ranges to collapse.

III. The ESG Siting Conflict: Coldwater Conservation vs. Big Tech

As tech giants (Microsoft, Google, Meta) and colocation operators (QTS Data Centers) face intense regulatory scrutiny, their hydrologic footprints have become a primary target for environmental groups and local communities:

1. The Trout Unlimited "Coldwater Conservation" Narrative

- Trout Unlimited is nationally recognized for protecting coldwater fisheries. The organization is highly vocal about industrial activities that cause stream temperature increases or aquifer drawdowns.
- **The Siting Risk:** If a data center developer sites a new campus in an exurban watershed that borders native Brook Trout headwaters (such as the Upper Toccoa or Etowah basins), local conservation groups can mount significant public opposition.
 - **The Reputational Risk:** A news headline stating "*Data Center Water Use Threatens Georgia's Last Native Spawning Riffles*" can severely impact a corporation's public ESG rating and delay multi-million-dollar zoning approvals for years.

IV. The Blue Ridge Solution: High-Value Spawning Sanctuary Offsets

To resolve this conflict and turn corporate risk into a highly positive conservation narrative, Blue Ridge Stream Restoration has structured a premium B2B **Water and Thermal Offset Framework**. Hunter Morris and Hadi Irvani present this strategic offset narrative directly to data center sustainability executives:

[DATA CENTER WATERSHED OFFSET VALUE CHAIN]		
Hyperscale Operator	Blue Ridge Spawning Sanctuary	Trout Unlimited & Warnell
+-----+	+-----+	+-----+
+ - Aquifer drawdown thermal risk	- Regan failing pasture banks => - 96% canopy solar attenuation	- Joint science research - Positive ESG ratings

	- ESG reputational		- Intercept cold seeps		- Fast-track regional
	vulnerability		- Create deep plunge pools		zoning approvals
	+-----+ +-----+ +-----+				
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The Sourcing Pitch and Strategic Alignment:

- 1. **The Thermal Offset Credit:** While a data center may consume groundwater in one part of a HUC-8 basin, they can *fully offset* this regional thermal impact by funding a **Blue Ridge Spawning Sanctuary** along a degraded reach in the same watershed.
- 2. **Turnkey Scientific Verification:** We install our off-grid **LoRa/Meshtastic water quality monitoring stations** on the funded reach. The live telemetry (water temperature, DO, stage) is streamed directly to Trout Unlimited and the UGA Warnell database, providing the corporate sponsor with real-time, mathematically verified evidence of temperature cooling and spawning success.
- 3. **Positive Corporate ESG PR:** Instead of a defensive public relations posture, the data center operator sponsors a high-status project with **UGA Warnell** and **Trout Unlimited**. They receive a branded, scientifically validated conservation legacy: *"This Coldwater Spawning Sanctuary was fully funded by [Sponsor Name] in partnership with Blue Ridge Stream Restoration to protect native wild brook trout."*
- 4. **USACE-Approved Mitigation Credits:** Sponsoring our Nationwide Permit 27 geomorphic restorations generates official Savannah District stream mitigation credits that operators can use to satisfy regulatory requirements for their new campus construction, creating a highly efficient, dual-value investment.

Developed by Blue Ridge Stream Restoration Technical Sourcing Operations. Pushed to remote origin under main.